2017-Winter Paper 1 – Basic Sciences in Urology

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Q1. Describe anatomy of pelvicalyceal system in detail and its application in endourological surgeries

Type: LAQ (25 marks)

Introduction

- Papilla → minor calyx → infundibulum → major calyx → renal pelvis → UPJ (ureteropelvic junction); lined by urothelium. Anatomy is very variable, so map it (CT (computed tomography) urogram or pyelogram) before every endourological procedure.

Anatomy of the calyces

- **Minor calyx:** a cup around one papilla; its two side projections are the **fornices**. **Compound calyces** (2–3 papillae) are commonest at the poles. Average 7–9 calyces per kidney (range 4–19 or more).
- **Distribution:** upper pole usually 3 calyces, interpolar 3–4, lower pole 2–3.
- **Drainage pattern:** upper pole by a single midline infundibulum; lower pole by a single midline infundibulum or paired calyces; hilar region by **anterior and posterior rows** of paired calyces.
- **Orientation:** the kidney lies on psoas, rotated 30–50° behind the coronal plane, so posterior calyces point posterolaterally (towards the prone surgeon). On a pyelogram anterior and posterior calyces overlap.
- **Brödel vs Hodson pattern (Kaye and Reinke):** in a Brödel-type kidney the posterior calyces lie lateral and the anterior calyces medial; in a Hodson-type kidney the reverse. Brödel type is commoner on the right, Hodson type on the left, so confirm calyx orientation with contrast/air or ultrasound before puncture.

Renal pelvis

- Pyramidal, apex to the ureter; capacity 3–10 mL. **Intrarenal** (in the sinus), **extrarenal** (larger, longer infundibula, mimics dilatation) or mixed.
- **Duplex (divided) pelvis:** upper and lower parts join at the hilum, with no direct connection between upper and lower calyces. Suspect it when one pole cannot be reached from a puncture.

Sampaio classification (140 resin endocasts)

Type	Frequency	Midzone (hilar) drainage
A-I	45%	Midzone calyces drain into the superior or inferior calyceal group

Type	Frequency	Midzone (hilar) drainage
A-II	17.2%	**Crossed calyces**: one to the superior and one to the inferior group
B-I	21.4%	Independent major calyceal group from the midzone
B-II	16.4%	1–4 minor calyces enter the pelvis directly

- **Perpendicular minor calyx (11.4%):** drains straight into the pelvis, hidden on a pyelogram; narrow neck (≤4 mm) hinders clearance.
- **Crossed calyces:** the limb to the inferior group is **ventral in 87.5%**.

Vascular relations (the basis of safe access)

- Hilum, anterior to posterior: vein, artery, pelvis, **posterior segmental artery** (the only branch passing behind the pelvis).
- Five segmental **end arteries**; interlobar arteries run in the columns of Bertin, hugging the anterior and posterior faces of the infundibula.
- **Brödel line:** avascular plane between anterior and posterior arterial territories, just posterior to the lateral border.

Puncture route (Sampaio)	Vascular injury
Upper-pole infundibulum	67% (arterial 26%); posterior segmental artery
Mid-pole infundibulum	Arterial 23%
Lower-pole infundibulum	Infundibular artery in 38%
Calyceal fornix	Venous <8%, **no arterial injury**

- **Relations:** pleura reaches the 12th rib posteriorly, lung the 10th intercostal space; liver and spleen descend in inspiration; **retrorenal colon** is commoner at the lower pole and when prone.

Applications in endourology

1. PCNL (percutaneous nephrolithotomy) access

- Ideal target: a **posterior calyx with a short, wide infundibulum**, punctured **through the papilla and fornix** along the calyx axis, never into the infundibulum or pelvis. This passes through a pyramid and avoids interlobar arteries.
- **Upper-pole posterior access** gives the straightest route to the pelvis, UPJ and most calyces (staghorn). Chest risk (pneumothorax, hydrothorax) rises above the 12th rib, highest above the 11th. Puncture in expiration.
- **Lower-pole access** is chest-safe but risks colon (left) and reaches the UPJ poorly.
- Identifying posterior calyces: rainbowing the C-arm towards the surgeon shortens them; injected air fills non-dependent (posterior, when prone) calyces; ultrasound also shows bowel and pleura.

2. RIRS (retrograde intrarenal surgery)

- Entering the lower-pole infundibulum needs about **140° deflection** (104–175°); ≥270° scopes see it, but working there is harder.
- **Unfavourable lower-pole anatomy:** steep infundibulopelvic angle, long calyx (>10 mm), narrow infundibulum (<5 mm). Relocate stones to an upper calyx with a basket.
- Lower-pole stone >1 cm: PCNL preferred (**Lower Pole I, 2001:** stone-free 91% PCNL vs 21% SWL (shock wave lithotripsy)).

3. Endopyelotomy

- Incise the UPJ **laterally**, where crossing vessels are absent. A crossing vessel lowers success (42% vs 86%).

- Antegrade route: mid-posterior or superolateral calyx; not lower pole (acute angle to the UPJ).

4. Infundibulotomy and calyceal diverticulum

- Incise infundibular stenosis **superiorly or inferiorly**; never anteriorly.
- **Calyceal diverticulum:** cortical cavity joined to the fornix by a narrow neck; treat if symptomatic. Upper or interpolar: ureteroscopy, incise or dilate the neck. Lower pole: direct puncture onto the stone, mini-PCNL; fulguration optional.

5. Open/anatrophic nephrolithotomy: incision along the Brödel line.

- **Also know:** horseshoe kidney (medial access, upper pole usually subcostal); pelvic kidney (ultrasound or laparoscopic guidance); megacalycosis mimics obstruction; AV (arteriovenous) fistula or pseudoaneurysm → angioembolisation.

Diagram to draw: transverse section of kidney showing anterior and posterior calyces, Brödel line and the needle through the posterior calyceal fornix.

> **Key points to remember:**
> - Sampaio: A-I 45%, A-II (crossed) 17.2%, B-I 21.4%, B-II 16.4%.
> - Fornix puncture: no arterial injury; upper-pole infundibulum 67% vascular injury.
> - Lower pole: steep angle, long calyx >10 mm, infundibulum <5 mm unfavourable.
> - Endopyelotomy lateral; infundibulotomy superior or inferior, never anterior.

Q2. Describe physiology of spermatogenesis and discuss management of azoospermia.

Type: LAQ (25 marks)

Hormonal control: HPG (hypothalamic–pituitary–gonadal) axis

- Hypothalamus releases **GnRH (gonadotropin-releasing hormone)** in pulses; pulsatility is essential (constant agonist exposure shuts the axis down). Kisspeptin triggers puberty.
- **LH (luteinising hormone)** → Leydig cells → testosterone (about 5 mg/day). **FSH (follicle-stimulating hormone)** → Sertoli cells.
- Feedback: testosterone (and oestradiol) on LH; **inhibin B** from Sertoli cells on FSH. So FSH and inhibin B report the tubules, LH reports Leydig cells.
- **Intratesticular testosterone** is about 50-fold higher than serum (held by **ABP (androgen-binding protein)**). This is why exogenous testosterone suppresses spermatogenesis.
- Testis is kept 2–4 °C below core temperature by the pampiniform countercurrent exchange.

Sertoli cells and blood–testis barrier

- Nurse cells: nourish germ cells, secrete ABP, inhibin, fluid; phagocytose residual bodies.
- **Sertoli–Sertoli tight junctions** divide the tubule into a **basal** compartment (spermatogonia) and an **adluminal** compartment (spermatocytes, spermatids). Post-meiotic cells are "foreign" to the immune system; the barrier and local tolerance protect them.

Phases of spermatogenesis (about 64 days; 42–76 days)

Phase	Events
1. Proliferation (spermatocytogenesis)	Dark type A (Ad, reserve stem cells) self-renew; pale type A (Ap) divide every 16 days into type B spermatogonia → primary spermatocytes. Daughter cells stay joined by cytoplasmic bridges
2. Meiosis	Primary spermatocytes cross into the adluminal compartment. Meiosis I (reduction division; crossing-over at synapsis) → secondary spermatocytes → meiosis II → 4 haploid round spermatids
3. Spermiogenesis	No division: acrosome from Golgi, flagellum from centriole, nucleus condenses to about 10%, mitochondria form the midpiece, cytoplasm shed
4. Spermiation	Sertoli cell releases sperm into the lumen

- **Cycle, stage, wave:** six stages in humans; cycles arranged as a helical wave, so production is continuous (about 123 million sperm a day).
- Defective meiotic recombination causes maturation arrest and sperm aneuploidy.

Epididymal maturation and transport

- Testicular sperm are immotile. During 2–12 days of epididymal transit they gain **progressive motility** and fertilising ability; stored in the cauda.
- **Capacitation** and the acrosome reaction occur in the female tract.
- **Clinical relevance:** a new semen sample reflects events about 3 months earlier.

Azoospermia: definition and classification

- No sperm in the **centrifuged** pellet on **two** semen analyses; 10–15% of infertile men.

Type	Causes	Clues
Pre-testicular	Kallmann syndrome, pituitary tumour, hyperprolactinaemia, **exogenous testosterone/anabolic steroids**	Low FSH, LH, testosterone
Testicular (NOA (non-obstructive azoospermia))	Klinefelter, Y microdeletion, cryptorchidism, mumps, chemotherapy/radiation	Small testes, high FSH
Post-testicular (OA (obstructive azoospermia))	CBAVD (congenital bilateral absence of the vas), epididymal obstruction, vasectomy, hernia repair, EDO (ejaculatory duct obstruction)	Normal testes and FSH
Ejaculatory	Retrograde ejaculation, anejaculation	Low volume

Evaluation

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Azoospermia (2 centrifuged samples)
 ├─ Vasa absent → CBAVD → CFTR (cystic fibrosis gene) testing, both partners; renal scan
 ├─ Volume <1 mL → post-ejaculate urine: sperm = retrograde; none → TRUS (transrectal
ultrasound) for EDO
 └─ Normal volume: FSH ≤7.6 IU/L + testis long axis >4.6 cm → obstructive
                   FSH >7.6 + testis ≤4.6 cm → spermatogenic failure
```

- History (testosterone use, surgery, chemotherapy), examination (testes, vasa, epididymis, varicocele), hormones (FSH, LH, testosterone, prolactin).
- **Genetics (AUA/ASRM (American Urological Association/American Society for Reproductive Medicine)):** karyotype and **Y microdeletion** for NOA. AZF (azoospermia factor) **a or b complete deletion: no sperm, do not operate**; AZFc: sperm often found.
- TRUS: seminal vesicle >1.5 cm wide or midline cyst suggests EDO.

Management

1. Pre-testicular (hypogonadotropic)

- Stop exogenous testosterone. **hCG (human chorionic gonadotropin) 1500–5000 IU 2–3 times weekly ± FSH**; pulsatile GnRH in hypothalamic failure; dopamine agonist for prolactinoma.

2. Obstructive

- Microsurgical **vasovasostomy** or **vasoepididymostomy** (clear watery fluid or sperm parts in vasal fluid → vasovasostomy; thick toothpaste-like fluid without sperm → vasoepididymostomy).
- **TURED (transurethral resection of the ejaculatory ducts)** for EDO.
- CBAVD or unreconstructable: **MESA (microsurgical epididymal sperm aspiration)** or PESA (percutaneous epididymal sperm aspiration) + **ICSI (intracytoplasmic sperm injection)**; freeze sperm at reconstruction.

3. Non-obstructive

- **Micro-TESE (microdissection testicular sperm extraction)** + ICSI. Spermatogenesis in NOA is focal; dilated opaque tubules are chosen under the microscope. Sperm found in about half (Klinefelter 68%, Sertoli-cell-only 30%).
- Before retrieval: repair a clinical varicocele; optimise hormones (clomiphene, anastrozole, hCG; off-label).

4. Ejaculatory

- Retrograde: sympathomimetics or sperm from alkalinised urine. Anejaculation: vibratory or electroejaculation.

5. Failure: donor sperm or adoption; genetic counselling.

- **Also know:** spermatogonial stem cell renewal by GDNF (glial cell line–derived neurotrophic factor); in-vitro spermatogenesis and stem-cell transplantation are experimental.

Diagram to draw: cross-section of a seminiferous tubule showing basal and adluminal compartments with germ cells maturing towards the lumen.

Key points to remember:

- Spermatogenesis about 64 days; Ap divide every 16 days; 6 stages.
- Intratesticular testosterone 50 × serum; exogenous testosterone causes azoospermia.
- FSH ≤7.6 IU/L and testis >4.6 cm → obstructive.
- AZFa/b deletion: do not operate; micro-TESE finds sperm in about half of NOA.

Q3a. Management of Peyronie's disease

Type: SAQ (10 marks)

Definition

- **PD (Peyronie's disease)** (AUA (American Urological Association) 2015): acquired fibrosis of the tunica albuginea, with pain, deformity, ED (erectile dysfunction) and/or distress. Prevalence 3–20%; up to half worsen untreated.

Phases

- **Acute (active, first 12–18 months):** pain, worsening deformity; aim to stop progression. **Stable:** pain gone, deformity unchanged for ≥3–6 months; aim to straighten.

Evaluation

- History: pain, deformity, Dupuytren disease, ability to penetrate. **Key question:** "If your penis were straight, would your current rigidity allow intercourse?" (predicts postoperative function).
- Plaque, SPL (stretched penile length); **office intracavernosal injection ± duplex Doppler ultrasound** before any invasive treatment (curvature by goniometer, calcification, vascular status).
- **Kelami grading:** ≤30° mild, 31–60° moderate, >60° severe; note hourglass or hinge. PDQ (Peyronie's Disease Questionnaire).

Non-surgical treatment

Option	Evidence/position
NSAIDs (non-steroidal anti-inflammatory drugs)	For pain (only oral agent AUA supports)
Oral vitamin E, tamoxifen, potaba, pentoxifylline	No reliable curvature benefit; not recommended
CCH (collagenase *Clostridium histolyticum*)	Stable disease, curve 30–90°, intact erections. **IMPRESS I/II:** 34% improvement vs 18% placebo. Up to 4 cycles (2 injections of 0.58 mg) with modelling. Risk: corporal fracture
Verapamil, interferon α-2b (intralesional)	May offer; stabilise and modestly improve
Traction therapy	Gains length, reduces curvature; useful in acute phase
ESWT (extracorporeal shock wave therapy)	Pain only; no effect on curvature

Surgery

- **Indications:** stable disease (≥12 months, stable ≥6 months), deformity preventing intercourse, failed conservative therapy.

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Erection adequate (± PDE5I (phosphodiesterase-5 inhibitor))?
 ├─ No  → penile prosthesis (inflatable) + manual modelling;
 │        residual >30° → tunical incision; graft if defect >2 cm
 └─ Yes → curve <60–70°, no hourglass/hinge → plication (Nesbit, 16-dot, tunica albuginea
plication)
          curve >60–70°, hourglass, hinge, short penis → plaque incision/partial
          excision + graft (pericardium, small intestinal submucosa)
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- **Plication** shortens the convex side: simple, low ED risk, but loses length.

- **Grafting** restores length and girth but carries higher ED risk (5–54%).

Diagram to draw: Nesbit wedge excised opposite the plaque.

> **Key points to remember:**
> - Ask the "straight penis" question; duplex before surgery.
> - CCH: IMPRESS, stable 30–90° curve, intact erections.
> - Surgery: stable ≥12 months; plication <60–70°, graft complex curves, prosthesis if ED.

Q3b. Treatment of renal abscess

Type: SAQ (10 marks)

Definition and types

- **Renal (cortical/corticomedullary) abscess:** pus confined to the renal parenchyma.
  - **Cortical (renal carbuncle):** haematogenous *Staphylococcus aureus* from a skin source; formerly about 80% of cases.
  - **Corticomedullary (now commonest):** ascending Gram-negative infection (*E. coli*, *Klebsiella*, *Proteus*) in a damaged or obstructed kidney (stones, reflux, diabetes).
- **Perinephric abscess:** outside the capsule, within Gerota fascia; paranephric beyond it.

Diagnosis (needed to plan treatment)

- Fever, flank pain, leucocytosis; urine may be sterile (haematogenous or non-communicating abscess). Blood cultures positive in a quarter to a third.
- **CT (computed tomography): investigation of choice:** round low-attenuation lesion with enhancing "ring sign". Ultrasound: quick, guides drainage. AFBN (acute focal bacterial nephritis) is wedge-shaped without a wall and is treated medically.

Treatment

1. Antibiotics immediately (after blood and urine cultures)

Presumed source	Empirical therapy
Haematogenous (Gram-positive)	Penicillinase-resistant penicillin; vancomycin if allergic or MRSA (methicillin-resistant *S. aureus*)
Ascending (Gram-negative)	IV (intravenous) third-generation cephalosporin, antipseudomonal penicillin or aminoglycoside

- Switch to oral once afebrile; continue until clinical and imaging resolution.

2. Size-based drainage

Size	Management
≤3 cm	Antibiotics alone (stable patient)
3–5 cm	Antibiotics alone or drainage; no consistent pathway
>5 cm	**Percutaneous drainage** (CT/ultrasound-guided) first line; may need several drains
Any size: immunocompromised or not responding	Percutaneous drainage

- Aspiration also distinguishes abscess from a cystic tumour; culture the pus.
- Serial ultrasound or CT until resolution. Worsening suggests perinephric spread, wrong diagnosis or resistant organism.

3. Treat the cause

- Relieve obstruction (stent or nephrostomy); treat stones once infection settles; control diabetes.

4. Surgery

- Open drainage for loculated or failed percutaneous drainage; **nephrectomy** for a non-functioning, destroyed kidney.

Perinephric abscess (if present)

- Lower threshold: drain if >3 cm; aminoglycoside + antistaphylococcal penicillin.

Diagram to draw: coronal kidney showing renal, perinephric and paranephric abscess sites.

Key points to remember:

- Now mostly ascending Gram-negative abscess in a damaged kidney.
- ≤3 cm antibiotics; 3–5 cm either; >5 cm percutaneous drainage.
- CT is the investigation of choice; follow until resolution.

Q3c. Catheter Associated Urinary Tract Infections (CAUTI)

Type: SAQ (10 marks)

Definition

- **CAUTI (catheter-associated urinary tract infection):** symptomatic UTI (urinary tract infection) with a catheter in place >2 days (or removed within 48 hours). Distinguish from **asymptomatic bacteriuria**, which needs no treatment.

Epidemiology and risk factors

- Commonest hospital-acquired infection (about 75–80% of hospital UTIs); commonest cause of secondary bacteraemia.
- Bacteriuria develops at about **5–10% per catheter-day**; 100% with long-term catheters. Intermittent catheterisation: 1–3% per catheterisation.
- **Risk factors:** duration (most important), female sex, older age, diabetes, breaks in the closed system, no systemic antibiotics.

Pathogenesis

- Entry at insertion, then **extraluminal** (along the outer surface; commoner in women) or **intraluminal** (from the drainage bag).
- **Biofilm:** bacteria in a matrix on the catheter surface, formed within minutes; it shields them from antibiotics. Urease-producers (*Proteus*) cause encrustation.
- Organisms: *E. coli*, *Pseudomonas*, *Proteus*, *Enterococcus*; long-term catheters polymicrobial.

Diagnosis

- Fever, rigors, flank or suprapubic pain, or delirium with no other source. Pyuria does not distinguish infection.
- Culture from a **freshly replaced catheter**, not the bag. **IDSA (Infectious Diseases Society of America): ≥10^3 CFU/mL** (colony-forming units) with symptoms; surveillance (CDC (Centers for Disease Control and Prevention)) definition uses ≥10^5.

Prevention (bundle)

- **Avoid unnecessary catheters;** review daily, remove early. Aseptic insertion.
- **Closed, dependent drainage**; bag below bladder; no routine irrigation.
- Meatal cleaning with water; no prophylactic antibiotics (delay bacteriuria only 3–4 days, select resistance).
- Alternatives: **clean intermittent catheterisation**, condom or suprapubic catheter.
- Silver-alloy catheters reduce bacteriuria in short-term use; no proven fall in symptomatic CAUTI.
- Do not treat asymptomatic bacteriuria (except pregnancy, before urological surgery).

Treatment

- **Replace or remove the catheter** before taking the culture and starting antibiotics, if in place >2 weeks (biofilm source).
- Empirical, then culture-directed antibiotics: **7 days** if prompt response; **10–14 days** if delayed.

Diagram to draw: catheter in bladder showing extraluminal and intraluminal routes and biofilm.

Key points to remember:

- Bacteriuria 5–10% per catheter-day; duration is the key risk.
- Diagnosis: symptoms + ≥10^3 CFU/mL from a fresh catheter.
- Remove catheter early; closed drainage; treat only symptomatic CAUTI, 7 or 10–14 days.

Q3d. Contrast Induced Nephropathy (CIN)

Type: SAQ (10 marks)

Definitions

- **CIN (contrast-induced nephropathy)** (older term): creatinine rise ≥25% or ≥0.5 mg/dL within 48–72 hours of iodinated contrast, with no other cause.
- Current terms (ACR (American College of Radiology)):
  - **CA-AKI (contrast-associated acute kidney injury):** AKI within 48 hours of contrast, whatever the cause.
  - **CI-AKI (contrast-induced AKI):** the subset actually caused by contrast.
- **KDIGO (Kidney Disease: Improving Global Outcomes) AKI:** creatinine rise ≥0.3 mg/dL in 48 hours, ≥1.5 × baseline in 7 days, or urine <0.5 mL/kg/h for 6 hours.

Pathophysiology

- **Renal vasoconstriction** causes **medullary hypoxia** and ischaemic tubular injury, plus **direct tubular toxicity**. Usually transient; dialysis rarely needed.

Risk factors

- **Patient:** CKD (chronic kidney disease, strongest), diabetes, age, dehydration, heart failure, anaemia, NSAIDs (non-steroidal anti-inflammatory drugs).
- **Procedure:** high volume, repeated doses within 48–72 hours, intra-arterial route, high-osmolar agents.
- **eGFR (estimated glomerular filtration rate) thresholds:** real risk mainly at **eGFR <30 mL/min/1.73 m²** (IV (intravenous) contrast); ESUR (European Society of Urogenital Radiology) uses **<45** for intra-arterial first-pass exposure.
- Older uncontrolled studies overstated the risk; ACR: rare but real.

Prevention

- Consider ultrasound or non-contrast CT (computed tomography) instead.
- Lowest dose of **low- or iso-osmolar** agent; low kVp (80); avoid repeat doses within 48 hours; stop NSAIDs.
- **IV isotonic saline** (e.g. 1 mL/kg/h before and after) for high-risk patients (eGFR <30 not on dialysis): the only proven measure.
- **PRESERVE (2018):** sodium bicarbonate and **NAC (N-acetylcysteine)** gave no benefit over saline.
- **AMACING (2017):** no prophylaxis was non-inferior to hydration at eGFR 30–59.
- Prophylactic dialysis does not help.

Metformin

- Risk is lactic acidosis if AKI develops. **eGFR ≥30 and no AKI: continue.** eGFR <30 or AKI: **withhold for 48 hours**, restart when creatinine is stable.

Diagram to draw: contrast → vasoconstriction + tubular toxicity → medullary hypoxia → AKI.

Key points to remember:

- CIN = rise ≥25% or 0.5 mg/dL in 48–72 h; now CA-AKI vs CI-AKI.
- Main risk: eGFR <30 (IV); <45 intra-arterial.
- Isotonic saline only proven; PRESERVE: bicarbonate and NAC useless.
- Metformin: withhold 48 h only if eGFR <30 or AKI.

Q3e. Transrectal ultrasound study of prostate

Type: SAQ (10 marks)

Principle and technique

- **TRUS (transrectal ultrasound):** the prostate lies directly in front of the rectum. Probes: end-fire, side-fire or biplane, **6–10 MHz** (higher frequency: better resolution, less depth). Left lateral (or lithotomy for transperineal work), after DRE (digital rectal examination). Transverse and sagittal planes, base to apex.

Normal sonographic anatomy

Zone (McNeal)	Glandular share	TRUS appearance	Cancer share
Peripheral	~70%	Homogeneous, echogenic (posterior)	~70%
Central	~25%	Echogenic, at base	~5–10%
Transition	5–10% (BPH (benign prostatic hyperplasia))	Heterogeneous, anterior	~20–25%
Anterior fibromuscular stroma	Non-glandular	Hypoechoic	–

- Zones are not seen as such; TRUS separates echogenic posterior PZ (peripheral zone) from heterogeneous TZ (transition zone). **Corpora amylacea** mark the surgical capsule.
- Seminal vesicles 4.5–5.5 × 2 cm; absent seminal vesicle → ipsilateral renal agenesis in 79%.

Measurements

- **Volume (ellipsoid) = 0.52 (π/6) × transverse × AP (anteroposterior) × length**; 1 mL ≈ 1 g; normal 20–25 g. **Planimetry** (serial sections) for brachytherapy.
- **PSAD (prostate-specific antigen density)** = PSA/volume; **>0.15** suggests cancer.

Findings

- **Cancer:** classically a hypoechoic PZ lesion, but **39% are isoechoic**; false positives: BPH nodules, prostatitis, infarct.
- **Cysts:** Müllerian (midline), ejaculatory duct cyst (just off midline), seminal vesicle cyst (lateral, renal agenesis).
- Abscess, calculi, extraprostatic extension.
- After ADT (androgen deprivation therapy) or radiotherapy the gland shrinks and becomes hypoechoic.

Indications

- **Guided biopsy** (12-core systematic ± MRI (magnetic resonance imaging)-targeted fusion).
- Volume before BPH surgery, brachytherapy, cryotherapy; fiducial markers.
- **Azoospermia:** ejaculatory duct obstruction (seminal vesicle >1.5 cm wide, midline cyst).
- Drainage of prostatic abscess or cysts.

Biopsy notes

- Periprostatic block. Complications: bleeding, **sepsis** (fluoroquinolone-resistant *E. coli*); transperineal route lowers infection.

Advances

- Colour/power Doppler, contrast-enhanced TRUS, **elastography**, multiparametric ultrasound.
- **Micro-ultrasound (29 MHz;** 70 µm resolution) with PRI-MUS scoring, comparable to MRI.
- MRI-first pathway: PROMIS (2017) mpMRI (multiparametric MRI) sensitivity 93% vs 48% for TRUS biopsy.

Diagram to draw: transverse and sagittal TRUS views labelling PZ, TZ, seminal vesicles and urethra.

> **Key points to remember:**
>
> - 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15.
> - 39% of cancers isoechoic; hypoechoic PZ lesion classic.
> - Seminal vesicle >1.5 cm wide: ejaculatory duct obstruction.

Q3f. Oncogenes

Type: SAQ (10 marks)

Definition

- **Proto-oncogene:** a normal gene that promotes cell growth and division (an "accelerator").
- **Oncogene:** its mutated or overactive form that drives uncontrolled proliferation. Change is **gain of function** and **dominant**: one altered allele is enough.
- The first human oncogene, **HRAS**, was isolated from the **T24 bladder cancer** cell line.

Mechanisms of activation

Mechanism	Effect	Example
Point (activating) mutation	Constitutively active protein	RAS (bladder), FGFR3 (fibroblast growth factor receptor 3; low-grade papillary bladder), MET germline (hereditary papillary RCC (renal cell carcinoma))
Gene amplification	Increased copy number → overexpression	MYC (prostate, bladder), HER2 (human epidermal growth factor receptor 2), AR (androgen receptor) in castration-resistant prostate cancer
Chromosomal translocation/fusion	Gene placed under a strong promoter, or fusion protein	**TMPRSS2–ERG** (androgen-driven ERG in about half of prostate cancers); BCR–ABL; TFE3 fusions (translocation RCC)

Functional classes

Class	Example
Growth factors	PDGF (platelet-derived growth factor), sis
Receptor tyrosine kinases	EGFR (epidermal growth factor receptor), HER2, FGFR3, MET, KIT (seminoma)
Signal transducers	**RAS** (binds GTP (guanosine triphosphate); mutation locks it "on"), BRAF, PI3K
Transcription factors	**MYC**, ERG, ETS family
Cell cycle / apoptosis regulators	Cyclin D1, BCL-2

Oncogene versus tumour suppressor gene

	Oncogene	Tumour suppressor
Normal role	Accelerator	Brake, DNA repair
Change	Gain of function	Loss of function
Alleles needed	One	Usually both (**Knudson two-hit**: retinoblastoma)
GU (genitourinary) examples	MYC, MET, FGFR3, ERG	RB1, TP53, VHL, PTEN

Urological significance

- **Bladder:** FGFR3 mutations in up to 80% of low-grade Ta tumours (10–20% of MIBC (muscle-invasive bladder cancer)); RAS mutually exclusive with FGFR3; HER2 in 9–12% of MIBC.
- **Kidney:** MET activation in papillary RCC.
- **Prostate:** TMPRSS2–ERG fusion, MYC (8q gain), AR amplification after hormone therapy.
- **Testis:** KIT and KRAS mutations (seminoma).

Targeted therapy (oncogene addiction)

- Erdafitinib (pan-FGFR inhibitor) for FGFR-altered advanced urothelial cancer; MET-active TKIs (tyrosine kinase inhibitors) in papillary RCC; AR-targeted drugs in prostate cancer; HER2-directed drugs (trastuzumab deruxtecan received a tumour-agnostic FDA approval in 2024 for HER2-positive solid tumours; disitamab vedotin is under study in urothelial cancer).

Diagram to draw: growth factor → receptor → RAS → RAF → MEK → ERK → MYC pathway.

> **Key points to remember:**
>
> - Oncogenes: gain of function, one allele; activated by mutation, amplification, translocation.
> - HRAS first found in T24 bladder cancer; FGFR3 in up to 80% low-grade Ta.
> - TMPRSS2–ERG in about half of prostate cancers; MET in papillary RCC.

Rapid Revision Sheet – 2017-Winter Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe anatomy of pelvicalyceal system in detail and its application in endourological surgeries (LAQ)

- Sampaio: A-I 45%, A-II (crossed) 17.2%, B-I 21.4%, B-II 16.4%.
- Fornix puncture: no arterial injury; upper-pole infundibulum 67% vascular injury.
- Lower pole: steep angle, long calyx >10 mm, infundibulum <5 mm unfavourable.
- Endopyelotomy lateral; infundibulotomy superior or inferior, never anterior.

Q2. Describe physiology of spermatogenesis and discuss management of azoospermia. (LAQ)

- Spermatogenesis about 64 days; Ap divide every 16 days; 6 stages.
- Intratesticular testosterone 50 × serum; exogenous testosterone causes azoospermia.
- FSH ≤7.6 IU/L and testis >4.6 cm → obstructive.
- AZFa/b deletion: do not operate; micro-TESE finds sperm in about half of NOA.

Q3a. Management of Peyronie's disease (SAQ)

- Ask the "straight penis" question; duplex before surgery.
- CCH: IMPRESS, stable 30–90° curve, intact erections.
- Surgery: stable ≥12 months; plication <60–70°, graft complex curves, prosthesis if ED.

Q3b. Treatment of renal abscess (SAQ)

- Now mostly ascending Gram-negative abscess in a damaged kidney.
- ≤3 cm antibiotics; 3–5 cm either; >5 cm percutaneous drainage.
- CT is the investigation of choice; follow until resolution.

Q3c. Catheter Associated Urinary Tract Infections (CAUTI) (SAQ)

- Bacteriuria 5–10% per catheter-day; duration is the key risk.
- Diagnosis: symptoms + ≥10^3 CFU/mL from a fresh catheter.
- Remove catheter early; closed drainage; treat only symptomatic CAUTI, 7 or 10–14 days.

Q3d. Contrast Induced Nephropathy (CIN) (SAQ)

- CIN = rise ≥25% or 0.5 mg/dL in 48–72 h; now CA-AKI vs CI-AKI.
- Main risk: eGFR <30 (IV); <45 intra-arterial.
- Isotonic saline only proven; PRESERVE: bicarbonate and NAC useless.
- Metformin: withhold 48 h only if eGFR <30 or AKI.

Q3e. Transrectal ultrasound study of prostate (SAQ)

- 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15.
- 39% of cancers isoechoic; hypoechoic PZ lesion classic.
- Seminal vesicle >1.5 cm wide: ejaculatory duct obstruction.

Q3f. Oncogenes (SAQ)

- Oncogenes: gain of function, one allele; activated by mutation, amplification, translocation.
- HRAS first found in T24 bladder cancer; FGFR3 in up to 80% low-grade Ta.
- TMPRSS2–ERG in about half of prostate cancers; MET in papillary RCC.